Tidal response across a coastal–inland optical-clock baseline

Working Article for collaboration review. Authors, affiliations and correspondence require confirmation.

Optical clocks make gravitational potential accessible through frequency comparisons between distant sites.[1–3] A coastal–inland baseline samples the combined response to astronomical forcing and ocean loading, with lunar phase evolution helping distinguish tides from solar-day variations. Here we analyse independently acquired strontium–ytterbium comparisons over the 1,350-km Shanghai–Wuhan fibre route.[4] Six observation periods in 2026 provide 20.1 days of retained sample time across a 231-day span. The scale of the predicted tide is 0.88 with a working standard error of 0.10 under ten-day calendar baselines; a period-resampling 95% acceptance interval is 0.51–1.42. Amplitudes estimated from other periods improve held-out residuals in all six periods. We estimate in-phase and quadrature coefficients of a common lunar response and assess their sensitivity to free solar-day terms. The lunar estimate remains compatible with the prediction, although period resampling limits separate attribution. A simultaneous round-trip monitor produces only a small change in the whole-template scale. These results complement contemporaneous independent network measurements[5] by establishing a tidal-response benchmark for interrupted operation and showing how accumulated exposure and lunar-phase coverage together determine measurement capability.

The gravitational redshift makes a clock's rate a local probe of gravitational potential. Optical-clock development has brought this principle into precision measurement,[6] from laboratory redshift tests[7] to resolving a potential gradient within a millimetre atomic sample.[8] Coherent networks[1,9] and transportable clocks[2] extend comparisons between locations. The Moon and Sun, deformation of the Earth and redistribution of mass produce time-variable signals on top of the static potential difference.[10,11] Observing these variations connects frequency metrology to dynamic geodesy.

A coastal–inland comparison is sensitive to both solid-Earth tides and the geographically varying ocean-loading response. It also offers a time-domain discriminator: the dominant lunar semidiurnal component M2 has a period of about 12.42 h, so its maxima move through local solar time by roughly 50 min per day. This phase evolution can separate a lunar response from variations tied to the laboratory day. Interrupted observations retain that information, but its usefulness depends on when samples are acquired and on how freely instrumental baselines and solar-day terms are allowed to vary.

We examine these questions with the Shanghai–Wuhan comparison. The principal result is a common response to the full predicted tide across six observation periods, tested by transferring an amplitude learned in other periods. We then separate lunar amplitude and phase, quantify capability in accumulated sample time, and test the simultaneously monitored transfer channel. Our earlier ratio preprint reported a tidal correlation in the summer record.[4] The present reconstruction distinguishes that result from the additional targets studied here (Table 1). The contemporaneous European experiment tracks tidal variations with a network and Kalman estimation;[5] together the two experiments address different operating conditions for dynamic clock comparisons.

A coastal–inland clock comparison

The comparison links a ^{87}Sr optical lattice clock in Shanghai with a ^{171}Yb optical lattice clock in Wuhan. The associated Sr1 evaluation reports a systematic uncertainty of 9.2×10^{-19},[12] and the Yb evaluation reports 1.3×10^{-18}.[13] These apparatus evaluations document the clock technology; period-specific corrections for the present observations follow the experimental ledger (Methods). The 1,350-km commercial-fibre route uses relay stations and bidirectional optical amplification.[4] Recent field-link demonstrations[14,15] provide further technical context; transfer sensitivity here is evaluated using the simultaneous record from this experiment.

The primary data comprise 1,737,073 retained source measurements in 2,905 non-overlapping 600-s averages, from 49 effective intervals between 7 January and 26 August 2026. Six periods cover January, April, May, June–July, and early and late August, with measurements on 35 BJT calendar days. Their 20.1 nominal sample days measure accumulated exposure within a 231-day span. Ratios are reconstructed from the recorded frequency chain and correction ledger, retaining interval levels relative to one numerical reference. The 2025 commissioning measurements, acquired under different conversion and processing conventions with unresolved offsets, are analysed separately within September and October; they do not enter the common 2026 amplitude.

The tidal prediction combines solid-Earth and ocean-loading contributions. On the actual retained

sampling, their demeaned root-mean-square amplitudes are 3.245×10^{-18} and 1.507×10^{-18}, with a combined amplitude of 4.410×10^{-18}. The loading-to-total RMS ratio is 34.2%; the contributions are correlated, so this is not a variance fraction. These are predicted contributions, rather than separate measurements of Earth and ocean response. Matched source-member averaging ensures that gaps and sampling affect the observations and prediction consistently (Methods).

A common response across observation periods

We fit a signed scale A_T multiplying the entire predicted fractional-frequency tide $T(t)$,

$$y(t) = A_T T(t) + \gamma_{b(t)} + \epsilon(t),$$

where $b(t)$ assigns observations to fixed ten-day BJT calendar blocks, ten of which contain data. The scale and baselines are estimated jointly. Measured points are not shifted interval by interval to align them with the prediction, and these short-period baselines are not used for seasonal inference.

The whole-template result is $A_T = 0.88 \pm 0.10$, where the uncertainty is the largest of the least-squares, delete-day and delete-period standard errors. Five-day baselines give 0.886 ± 0.101, and independent linear drifts within ten-day blocks give 0.878 ± 0.101. The six-period restricted wild test gives a two-sided p-value of 0.03125 for the zero-response null, its finite-resolution minimum, and a 95% acceptance interval of [0.51, 1.42]. Thus the data support a response near the predicted scale within this baseline model, with uncertainty assessed at both day and period levels (Methods).

Figure 1a shows the 48-h window with the greatest retained source count. The plotted points retain their measured levels after one common reference subtraction. The displayed template uses an amplitude of 0.830 learned from the other five periods; only the prescribed calendar-block baselines are fitted locally. Across all six held-out periods, training amplitudes range from 0.830 to 0.965 and improve the residual sum of squares in every period, by 3.26% when pooled (Fig. 1c). This tests transfer of a common amplitude across interruptions; the folds share training data.

The phase-sensitive harmonic diagnostic provides a complementary view of the same record. Subtracting the fixed unit-amplitude, fixed-phase tide lowers the M2-frequency ordinate from 13.02×10^{-36} to 0.080×10^{-36}, a 99.4% reduction, under the same nuisance projection (Fig. 1b). This subtraction estimates neither amplitude nor phase from the observations. The ordinate is harmonic variance reduction per observation, rather than a physical power spectrum or a separate detection probability. Behaviour is frequency dependent: O1 increases after subtraction, as does N2 under daily-term controls; all specified frequencies are retained in Extended Data Table 1.

Separating lunar and solar-day variation

We next allow sine and cosine terms at exactly 24 h and 12 h within every occupied block. The whole-template scale becomes 0.79 ± 0.28 for ten-day baselines and 0.92 ± 0.38 for five-day baselines. On the ten-day projection, fixed-unit subtraction still lowers the M2 diagnostic by 91.8%. The free daily terms allow solar-day variation of unspecified origin; they do not assign it to a particular clock or environmental mechanism.

To isolate the lunar semidiurnal response, we estimate a common M2+N2 scale while allowing separate scales for the near-12-h and diurnal tidal groups. With the daily terms retained, $A_L = 0.89 \pm 0.28$ for ten-day baselines and 0.89 ± 0.43 for five-day baselines. Adding block-specific linear drifts gives 0.89 ± 0.29 and 0.82 ± 0.56, respectively (Fig. 2a). Three direct-potential alternatives change the lunar scale by less than 0.005 (Methods). These comparisons place the lunar response near the predicted scale across the specified controls.

Figure 2b makes the phase test explicit. The ten-day in-phase and quadrature coefficients are $(0.866, 0.104)$, without changing observation times. The unit, in-phase prediction lies within both nominal delete-day regions. Period-level inference is broader: the joint six-period test gives $p = 0.25$ for zero and $p = 0.72$ for the unit prediction. For the fixed-phase lunar scale, the ten-day 95% acceptance interval is $[-1.18, 4.22]$. Lunar amplitude and phase are therefore compatible with the prediction, while a separate zero lunar response is not robustly excluded by period resampling. This distinction is essential when interpreting the visually more restrictive delete-day ellipses.

Solar-day freedom also changes where the information resides. Summer contributes 75.8% of projected lunar-template squared norm, versus 45.9% for the baseline-only whole template. With daily terms, three of six held-out periods improve, by 0.467% in pooled residual sum of squares (Fig. 1c). The whole-template and lunar results consequently describe different statistical targets, rather than six independent repetitions of a lunar detection. Separate constituent fits give M2 and N2 scales of 0.94 ± 0.27 and 0.29 ± 1.08.

Accumulated measurement capability

Accumulated sample time connects the observed response to the capability of the comparison. At eight nominal sample days, whole-template and lunar working errors are 0.114 and 1.025; at 20.1 days they are 0.100 and 0.282. Figure 3 follows the same time-ordered record using delete-day errors and a fixed independent-bin benchmark based on the measured 600-s instability. The other error summaries are retained in Extended Data Table 3.

The gains reflect more than exposure alone. At four nominal days, the lunar benchmark error is 4.33 times an inverse-square-root extrapolation from full support, because the sampled phases and nuisance design differ. Later observations improve lunar separation as well as increasing exposure. This comparison identifies the roles of acquired sample time and astronomical phase coverage; it does not equate 20.1 accumulated days with uninterrupted stability or a universal detection time.

A simultaneous transfer control

The apparatus-designated B2 round-trip record provides an experimental check on a measured transfer channel. Unique source-member pairing, without interpolation and with the specified gross-excursion criterion, retains 1,519 complete bins in summer 2026. On simultaneous valid second samples, the clock comparison and optical-normalized B2 channel have one-hour overlapping Allan deviations of 7.66×10^{-18} and 6.26×10^{-18} (Fig. 4a).

Adding B2 as a regressor changes the whole-template scale by 0.0057 ± 0.0073 without daily terms and 0.0415 ± 0.0345 with them, using paired day deletions (Fig. 4b). B2's own normalized template projections are 0.074 ± 0.087 and 0.54 ± 0.46, respectively. The small change in the clock scale therefore constrains the measured linear association on simultaneous support. Threshold and pairing sensitivities, longer-averaging-time support and the optical normalization are specified in Methods. This control addresses the recorded channel; it does not exclude clock variations or unmonitored laboratory paths.

Discussion

The coastal–inland comparison supports a common response to a tidal model containing both solid-Earth and ocean-loading contributions. Its amplitude transfers across six separated periods, while the lunar phase analysis shows how the changing astronomical phase can be tested against solar-day freedom. The 20.1 nominal sample days quantify the exposure actually acquired, and the accumulation analysis connects that exposure to phase coverage and precision. These results provide an experimental benchmark for time-variable potential sensing under interrupted operation.

The distinction between short-period response and slower sensing is physical as well as statistical. Seasonal analyses preserve every measured interval level, fit one global intercept and never align segment centres (Extended Data Fig. 1). For this nearly equal-latitude baseline, the predicted semiannual solar response is only about 2×10^{-20}; even the no-drift conditional upper bound is about 5,500 times larger. Seasonal sensing therefore requires both calibrated levels and sufficient differential signal. The full long-period bounds, drift sensitivities and loading-model scope are retained in Methods.

Our earlier ratio preprint and this work share summer observations, but differ in data support, template construction, nuisance models and uncertainty procedures (Table 1). The earlier $0.87(15)$ and reported 5.7σ belong to that analysis and are not significance statements for the lunar component here. The present work adds component-level phase tests, transfer of amplitude across separated periods, and a direct accounting of measurement capability. Neither analysis independently separates ocean-loading or geophysical response parameters.

Stahl and colleagues demonstrate tidal tracking with a European network, Kalman estimation and same-site controls.[5] The Shanghai–Wuhan observations were acquired independently, according to the experimental team; the analyses developed here followed access to that work. The experiments offer comple-

mentary evidence across different baselines and operating conditions. Extending dynamic chronometric sensing will require maintaining the connection between astronomical phase, measured exposure and calibration, with clock and local-transfer monitoring strengthening attribution beyond the channel tested here.

References

1. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nature Communications* **7**, 12443 (2016). Source
2. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nature Physics* **14**, 437–441 (2018). Source
3. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). Source
4. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). Source
5. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). Source
6. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Reviews of Modern Physics* **87**, 637–701 (2015). Source
7. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). Source
8. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). Source
9. Riehle, F. Optical clock networks. *Nature Photonics* **11**, 25–31 (2017). Source
10. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). Source
11. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *Journal of Geodesy* **92**, 487–516 (2018). Source
12. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). Source
13. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). Source
14. Chen, F.-X. et al. 10^{-21}-Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications* **15**, 276 (2026). Source
15. Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nature Communications* **13**, 212 (2022). Source

Table 1

Table 1 | Related tidal estimates and their statistical targets. The estimates share observations and are not combined. Present common amplitudes use 2026 only, with 2025 commissioning records analysed separately. Values in parentheses or after ± denote the error scale specified in the last column; they are not a common calibrated confidence level.

Analysis	Observations and averaging	Tidal target and nuisance model	Scale	Error definition
Hou et al.[4]	Summer record; 16 intervals; 1,200-s triangular windows	Whole predicted tide; interval-centred, precision-weighted estimates; window roles distinguished below	0.87(15)	Reported weighted statistical error; reported 5.7σ under that analysis
Present 2026 reconstruction	2,905 non-overlapping 600-s means	Whole predicted tide; ten-day calendar baselines	0.88 ± 0.10	max(LS, delete-day, delete-period SE)
Present daily-term control	Same 2,905 means	Whole predicted tide; ten-day baselines and 24-h/12-h terms	0.79 ± 0.28	Same standard-error summary
Present lunar decomposition	Same 2,905 means	M2+N2; independent other tide groups and ten-day daily terms	0.89 ± 0.28	Same standard-error summary

The archived calculation uses 600-s-stride, 1,200-s triangular windows for amplitudes, but non-overlapping windows for standard errors and correlations, contrary to the common support stated in Hou et al.'s Table 8.[4] Using non-overlapping windows for amplitudes as well changes the precision-weighted scale from 0.87303 to 0.87756, with the same conditional standard error of 0.15193. This comparison does not recalibrate the reported significance.

On identical current summer observations (1,642 averages), ten-day baselines and common-amplitude least squares, the published and current templates give scales of 0.752 and 0.693. Historical-window comparisons also depend on precision weighting versus a common-amplitude fit. These controlled comparisons do not establish which template is physically more accurate or fully disentangle changes in selection, frequency conversion, timing and averaging. Extending from summer to all six periods additionally changes observing support; agreement with the earlier aggregate is not independent validation.

Figure legends

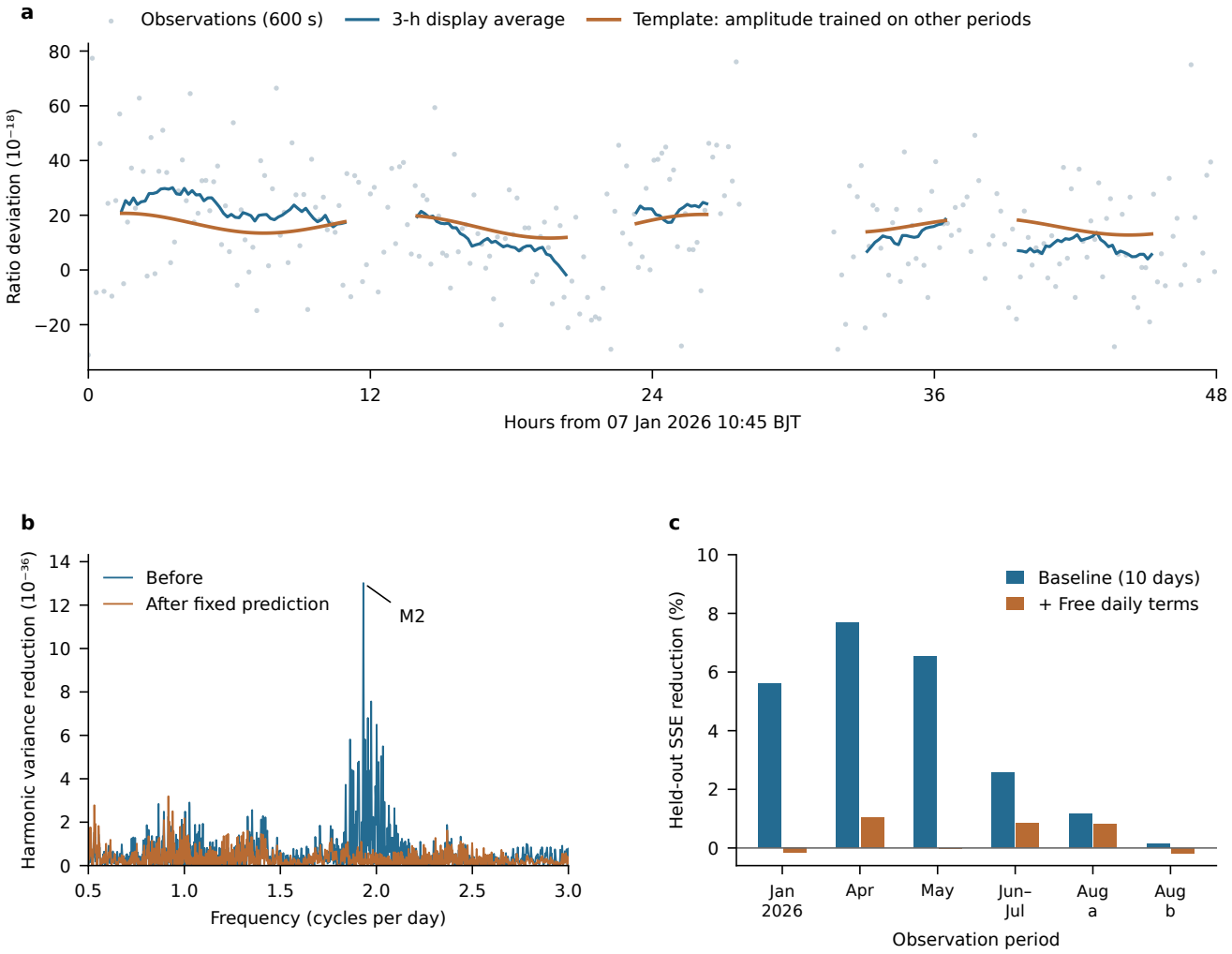


Fig. 1 | A common tidal-template response across interrupted observations. a, The 48-h window containing the largest number of retained source members (158,072 in 264 bins), selected without reference to fit quality. Points are 600-s averages after subtraction of one common reference used for the whole record. The blue curve averages 18 consecutive bins within uninterrupted supports for display. The orange curve applies the same display average to the template, with its amplitude trained on the other five periods and its calendar-block baselines fitted in the held-out period. No individual segment offsets are subtracted from the points. **b,** Actual-time harmonic scans of all 2,905 bins before and after subtracting the fixed unit-amplitude, fixed-phase prediction, with the same ten-day baseline projection. No amplitude or phase is fitted for this subtraction. The ordinate is harmonic variance reduction per observation, not a physical power spectral density or a significance threshold. **c,** Changes in held-out-period residual sums of squares when the amplitude is learned from the other periods. Blue and orange denote the baseline-only and free-daily-term designs. The held-out nuisance terms are fitted locally; the six folds share training data. All time-series points are repeated observations of the same apparatus, not independent experiments.

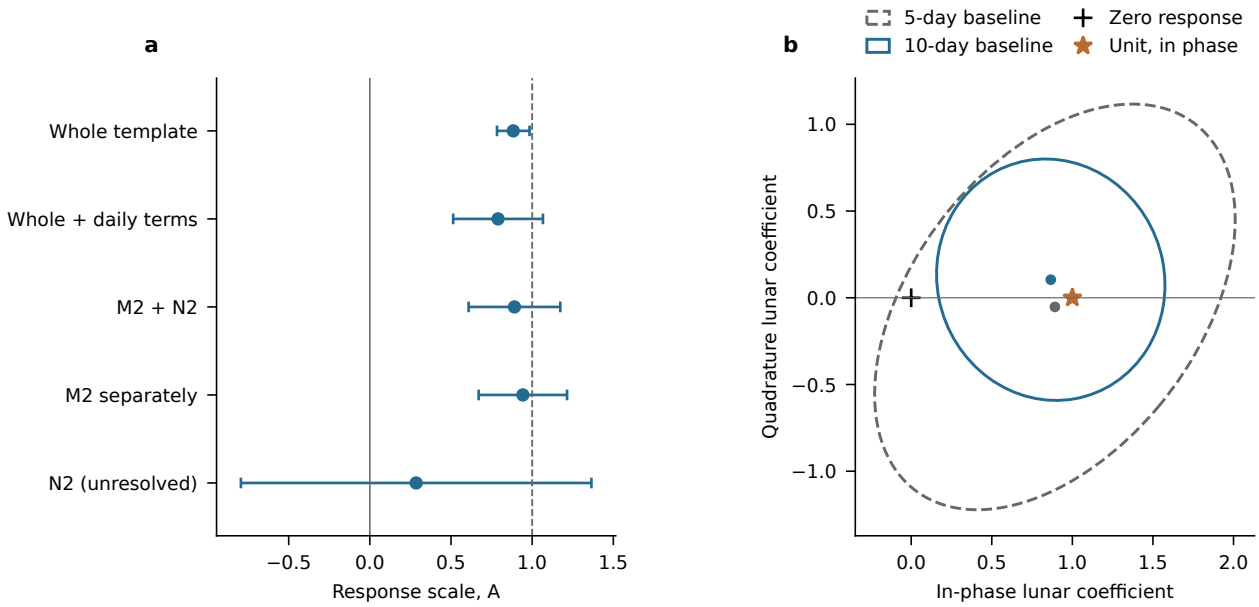


Fig. 2 | Lunar response after allowing solar-day variation. **a,** Scales estimated from the same 2,905 averages. All rows except the first include ten-day 24-h and 12-h nuisance terms. M2 and N2 are estimated jointly with independent scales in the last two rows. Bars show one standard-error summary, max(LS, delete-day, delete-period); the vertical references mark zero and unit response. The rows are correlated estimates of different targets. **b,** In-phase and quadrature coefficients of a common lunar response for five-day and ten-day baselines. Curves are nominal 95% delete-day working-model regions for the two coefficients; points mark their estimates. The cross and star denote zero and the unit, in-phase prediction. Zero is inside the five-day region and outside the ten-day region; the unit, in-phase point lies within both. No observation times or phases are adjusted. For the same two-parameter target, the six-period wild test gives ten-day $p = 0.25$ at zero and $p = 0.72$ at the unit point (Methods, Extended Data Table 2); it uses different covariance estimation and calibration from these ellipses. Neither procedure corrects for exploratory model selection or arbitrary noise.

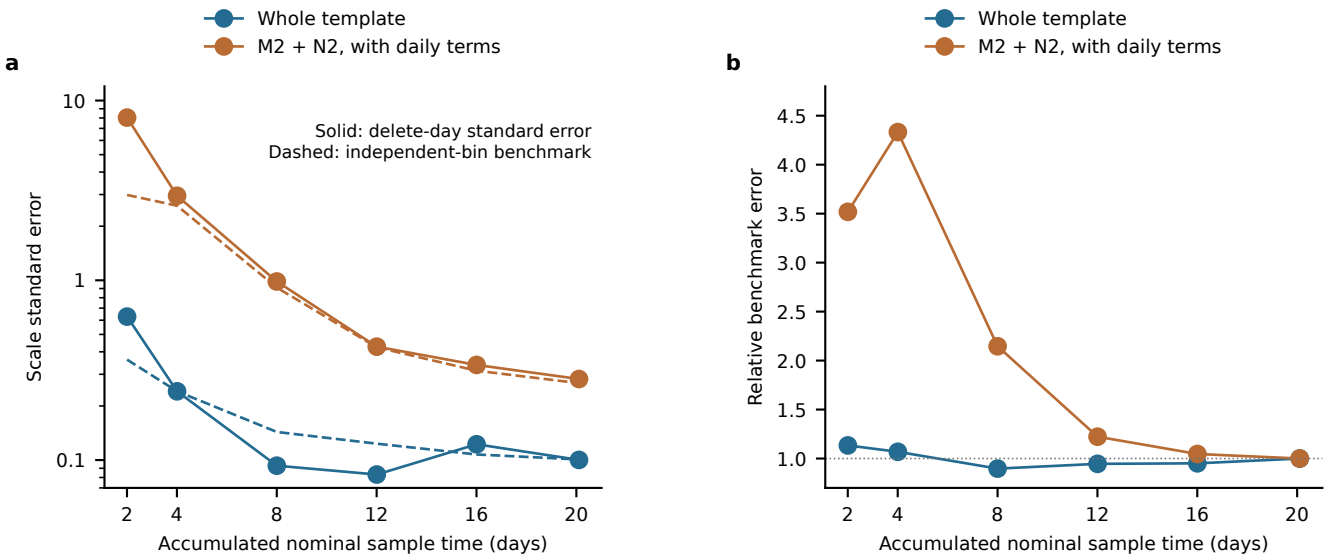


Fig. 3 | Accumulated sample time and conditional sensitivity. a, Six chronological prefixes of the 2026 record, ending at the first complete bin reaching 2, 4, 8, 12 or 16 nominal sample days, or at full support. The full record contains 2,905 means and 1,737,073 retained nominal one-second samples. Solid curves show delete-day standard errors of the whole-template and daily-controlled lunar scales; dashed curves show a fixed independent-bin benchmark based on the measured summer 600-s Allan deviation. Other error summaries are listed in Extended Data Table 3. **b,** Each benchmark error divided by its full-support value scaled by the inverse square root of sample time. This ratio describes sampling and nuisance-design differences. Colours identify the same targets in both panels. The prefixes overlap and change period and phase coverage; they are not independent replications. These pointwise errors do not define a universal detection time, and accumulated sample time does not establish continuous stability.

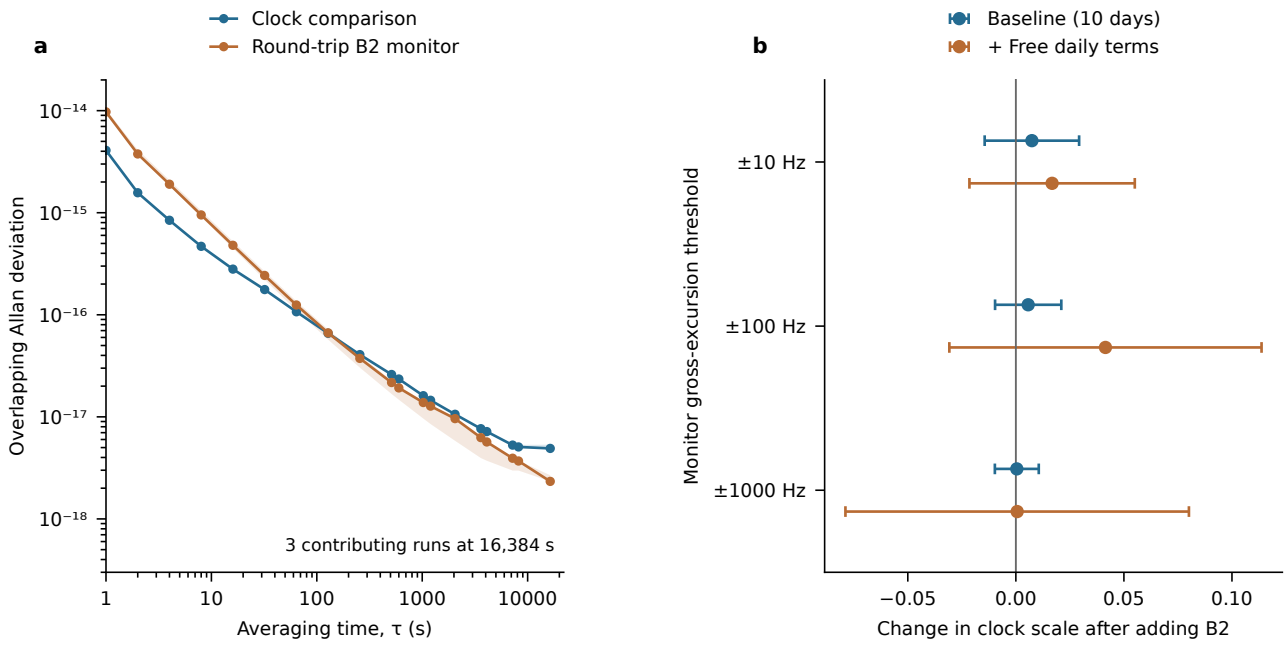


Fig. 4 | Scope of the measured round-trip transfer control. **a,** Overlapping Allan deviations of the clock comparison and predesignated B2 monitor on simultaneous valid second samples. Solid curves use the ±100-Hz gross-excursion criterion (913,692 common seconds before averaging-time run requirements). Shading spans the ±10-Hz and ±1000-Hz sensitivity results as well as the main curve; it is not a confidence band. No differences cross gaps, and contributing runs must span at least four averaging times. At 16,384 s, only three runs contribute 268,673 overlapping differences, which are not independent replicates. The monitor is normalized by the nominal optical carrier without an assumed one-way conversion. **b,** Paired changes in the clock's whole-template scale on adding B2, at zero nominal pairing shift. Thresholds leave 1,476, 1,519 and 1,520 complete 600-s bins. Error bars are nominal 95% delete-day intervals for the paired changes, with 19 degrees of freedom; they are not interval differences computed from independent fits. Baseline-only and free-daily-term designs use the same support within each threshold. The control covers the recorded channel and time overlap, including neither unmonitored laboratory branches nor the complete year.

Methods

Coordinate and frequency conversion

The principal inference uses 2026; the separately recorded 2025 conversion is retained for commissioning-stage diagnostics.[4] Let $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$. The 2025 chain gives

$$f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \quad \alpha = \frac{10.93629485}{25N_{1397}}, \quad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight Shanghai reference units include seven in the transmitted carrier and one added at Wuhan. The Wuhan repetition frequency is $f_{\mathrm{Yb}}/(2N_{1156})$, its reference is $f_{\mathrm{Yb}}/(40N_{1156})$, and its comb offset is 26 reference units. For recorded frequency q in nominal hertz realized by that Wuhan reference,

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}}\frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)}\frac{1}{1-\alpha}.$$

We use $d = 2$ and the stated α in 2025, versus $d = 1$, $\alpha = 0$ in 2026. The respective counter values are near 66 and 33.6 MHz; the former represents a 132-MHz beat before division. The counter-reference convention is an experimental assumption, not implied by a 10-MHz label. The electronic divider is restored once, independently of any round-trip correction.

For signed Sr and Yb shifts a and b, $R = R_{\mathrm{lock}}(1+a)/(1+b)(1+\Delta G)$, with inherited $\Delta G = -3.116\times 10^{-15}$. We form $y = (R/R_{\mathrm{ref}} - 1)10^{18}$ using $R_{\mathrm{ref}} = 1.2075070393433377166$ and Decimal anchors, avoiding subtraction of full ratios in float64. Positive y means faster Wuhan Yb relative to Shanghai Sr. For positive Newtonian potential U, the rate contribution is $-[U_{\mathrm{WUH}} - U_{\mathrm{SHA}}]/c^2$. These conventions do not establish a new absolute ratio or static potential.

The eight periods are September/October 2025, January/April/May 2026, June–July, and early/late August. In 10^{-18} units, Sr density corrections are [−0.43,−0.41,−0.30,−0.33,−0.25,−0.24,−0.33,−0.29], the first six Zeeman corrections are [−178.50,−178.87,−179.14,−180.19,−180.18,−178.57], and the background-gas term is −0.67. August Zeeman and all blackbody corrections are recorded as online. The first five AC entries are missing; the final three are [7.44,8.93,9.03]. Positive 37.2-mHz feedback applies to the first three periods at the assumed 698-nm optical node, giving 86.6119×10^{-18}—larger than the approximately 10×10^{-18} tide. Printed Yb shifts for original groups 1–9 are primary, with component sums tested separately; May Yb is missing. Computational zeros for unknown entries are not measured zeros. Missing terms have no certified bound. The ledger is frozen for these refits, but neither its precedence over all earlier residual inspection nor within-block constancy is established. Baselines absorb constants; time-dependent shifts can project onto the tide.

Commissioning records and corrections

Reported commissioning, processing changes and unresolved offsets motivate separate September and October 2025 stage- or interval-intercept analyses, without shared amplitudes across months or with 2026. This scope was clarified during reanalysis.

An older uncompensated Yb shift near -1.1×10^{-17} may overlap the updated totals -9.7 and -9.5×10^{-18}, which include a lattice entry of -8×10^{-18}. Inclusion is unverified; we do not add it twice. An additional δb multiplies R by $(1+b)/(1+b+\delta b)$, so $\delta b = -1.1\times 10^{-17}$ would increase R by approximately that fractional magnitude. Additive-shift invariance of intercept designs does not verify this multiplicative correction or its provenance. September and October stage fits give 0.87 ± 0.67 and 1.57 ± 0.26 (LS only), with larger errors after daily controls. Only two days per month preclude establishing correlated-noise coverage; these are commissioning diagnostics.

Observation selection and averaging

The corrected intervals select 1,966,912 unique source rows from 42 files, retaining fractional-second BJT labels. Windows include the first PDF second and exclude the second following the last. Rows are time/file/row sorted and deduplicated by source identity. Finite offsets below 1 MHz in magnitude and

strictly within 10 Hz of the plausible-sample median leave 1,966,908 rows. The three-row difference from the experimental count retains boundary provenance.

Within each interval, 300-s blocks start at the first source timestamp surviving the initial screen. Their means are compared with a 13-block running median with reflected boundaries. Empty blocks retain their fixed-grid positions as NaN means, ignored within each median window. Residual median absolute deviation divided by 0.6744897501960817 sets the scale. Blocks exceeding six scales and their neighbours are excluded, removing 11,131 rows without replacing retained values or using tidal residuals. Averaging instead starts at the floored earliest source timestamp before screening: half-open 600-s windows never cross effective intervals. Every window, including a partial tail, requires at least 300 retained members. The 3,266 means contain 1,952,992 members. Restricting to 2026 leaves 2,905 means and 1,737,073 members; September/October 2025 retain 160/201 means. Theory columns use identical members. Baseline indicators use mean bin times, retaining complete boundary-crossing bins; observations are not interpolated. Day deletions use mean-bin BJT days, not every raw sample of that civil day: the parent record includes 16 midnight-crossing bins. Retained 2026 samples occupy every local hour, with 45.6% between 06:00 and 18:00. Extended Data Table 5 lists ten-day support. No block crosses the year-convention change, and within-block drift tests identify no specific hardware event.

We repeated the fixed designs using no local screening and the retained six- and eight-scale rules. These leave 2,916, 2,905 and 2,909 bins, respectively, retaining the initial validity rules and minimum membership. Each scenario averages response and theory over identical members. Scalar and joint two-coefficient tests are separate; masks also change bin-mean times and design. The six-scale primary rule is unchanged. These correlated sensitivities neither identify physical lock failures nor calibrate screening-selection uncertainty.

Tidal prediction and component design

The common prediction combines solid-Earth and ocean-loading series on a 90-s grid. Pysolid displacements under the recorded IERS 2010 implementation[16] are converted using

$$s(t) = -\frac{1 + k_2 - h_2}{h_2}\frac{g[u_{\mathrm{WUH}}(t) - u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m\,s^{-2}}$. This displacement approximation is not a complete degree-dependent potential calculation. Loading uses 18 constituents from grids locally documented as FES2022b.[17] The FES2014 paper describes an earlier atlas, not validation of FES2022.[18] FES2022 was funded by CNES, produced by LEGOS, NOVELTIS and CLS, and made freely available by AVISO. Convolution uses PREM load-Love Green functions containing $1+k'_n-h'_n$; the Han–Wahr table[19] through degree 696 is extended to 40000, tapering the final 30%, with the recorded centre-of-solid-Earth degree-one convention. Direct water potential, induced potential and displacement are included. Member-matched solid, loading and total demeaned RMS values are $(3.245, 1.507, 4.410) \times 10^{-18}$. Loading/total RMS is 34.2%, with solid–loading correlation 0.680; variances are not additive. Atmospheric loading is neither modelled nor bounded. Free daily terms control empirical variation but do not validate an atmosphere-loading field, nor can local pressure differences do so.

The 360,001 theory samples span 1 September 2025, 08:00 to 11 September 2026, 08:00 BJT, inclusive. Equal-weight least squares over this full grid projects onto quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant (rank 27), retaining the remainder. BJT converts to UTC once, with the saved FES convention and TT−UTC=69.184 s. Harmonic columns, including the lunar quadrature, are evaluated on the 90-s grid and interpolated before source-member averaging. Whole-template fits scale the complete prediction; component fits subtract the remainder once. In 10^{-18} units,

$$y - F = A_L C_L + a_{12} C_{12} + a_D C_D + Z_b \gamma + Z_{24,12}\beta + \epsilon.$$

C_L, C_{12} and C_D contain M2+N2, S2+K2 and the four diurnal constituents; F contains remaining theory, including long-period terms. The daily design has sine/cosine columns at one and two cycles per day in each occupied block. The ten-day lunar design has 53 parameters and full rank. Blocks follow $\lfloor t/(10 \times 86400) \rfloor$ on the BJT epoch coordinate; five-day blocks follow the analogous rule. S2+K2 is not interpreted as resolved from exact 12-h terms. Drift tests add a within-block linear column, centred on observed mean time, on identical data; paired deletions estimate changes. These short-period baselines/slopes never enter seasonal fits.

Twenty-day, thirty-day and per-period controls give lunar scales 0.867 ± 0.246, 0.796 ± 0.202 and 0.761 ± 0.205 from the same sample time. Thirty-day blocks share a baseline/daily pattern across August periods; per-period models separate them. Their scalar wild $p = 0.03125$ does not motivate replacing the weaker five-/ten-day primary controls (joint tests: Extended Data Table 2).

Three sensitivities replace only the solid-Earth series with DE430 geometric potentials:[20] degree two with response $1+k_2-h_2$; degrees two/three with constant responses; and degrees two/three with nominal order-dependent real k_{nm} from IERS Table 6.3.[16] We use $h_3 = 0.292$, IAU 2006/2000A rotation, UT1≈UTC and zero polar motion. These first three alternatives hold the response coefficients constant in frequency. Whole-template scales become 0.876, 0.872 and 0.872; lunar scales become 0.895, 0.887 and 0.887. For the degree-two constant-response alternative, substituting Hou et al. coordinates[4] (Wuhan 30°31′ N, 114°19′ E, 30 m; Shanghai 31°14′ N, 121°29′ E, 5 m), with the same potential algorithm and loading, changes A_T by +0.0066 and A_L by +0.0046; solid-template difference RMS is 0.043×10^{-18}. Neither coordinate set is certified at the clock transition points; height references also require confirmation.

A separate finite-basis sensitivity applies nominal frequency-dependent $\gamma(\omega) = 1+k(\omega)-h(\omega)$ from the IERS resonance and anelastic approximations[16] implemented in pyTMD 3.0.9. We project the constant-response degree-two DE430 series onto the same 13 constituents, transform each pair by γ/γ_0, and retain the constant and residual from each projection and the original loading series. Here $\gamma_0 = 1 + 0.3010 - 0.6078$. Both projections use equal weights; transformed series are evaluated on the original full grid and averaged over the unchanged source members. For $C\cos\theta + S\sin\theta$ and $\gamma/\gamma_0 = a+ib$, the transformed coefficients are $(aC + bS, aS - bC)$. Relative to this same DE430 constant control, real-only and complex responses change the baseline-only whole-template scale by -0.00442 and -0.00434, respectively; changes in daily-controlled A_L are below 0.00008. Trimming 30 days from each end of the theoretical projection window changes these scales by less than 0.000007. The observations and reported primary estimates are unchanged; alternative estimates are retained as sensitivities. This nominal finite-basis check does not reproduce the full effective IERS response, separate nodal-sideband responses, latitude dependence or complete ocean feedback; its small shifts do not bound all theory errors.

Estimation, correlation and validation

Unweighted OLS uses column-normalized SVD. With n averages, $p = \mathrm{rank}(X)$, residuals r and the original-scale pseudoinverse $B = X^+$, $C_{\mathrm{LS}} = [\sum_i r_i^2/(n-p)]BB^T$. Delete-day and delete-period covariances are also retained. Long-period analyses additionally retain elapsed-time Bartlett HAC sensitivities at one, three and ten days. Jackknife covariance[21] is $C_{\mathrm{JK}} = (G-1)G^{-1}\sum_g(\hat{\theta}_{-g}-\bar{\theta})(\hat{\theta}_{-g}-\bar{\theta})^T$. Deletion-group independence remains an assumption. Unsupported nuisance columns are removed; target/rank loss is recorded. The maximum LS/day/period SE is a scalar summary, not a joint covariance. Two-coefficient regions use $(b-\hat{b})^T C^{-1}(b-\hat{b}) \le 2F_{2,\nu}(0.95)$, with $\nu = G-1$ for jackknife covariance and $n-p$ for LS. Figure 2b uses delete-day covariance with 34 denominator degrees of freedom.

Restricted wild tests[22] fit each null and enumerate 64 period-wise sign patterns, studentizing with delete-period covariance. Scalar tests use bootstrap-t; joint phase tests use both coefficients and their covariance (Extended Data Table 2). Period independence and error symmetry are working assumptions; nuisance estimation prevents exact randomization inference. The minimum two-sided p-value is 0.03125. Exploratory choices are not multiplicity-adjusted. Scalar inversion partitions the real line at polynomial roots, including endpoints and isolated-point checks. The ten-day daily-term whole-template 95% set is $[-0.541, -0.294] \cup [0.241, 0.367] \cup [0.385, 2.568]$, not its convex hull. Independent QR and covariance checks verify implementation agreement, not experimental coverage.

Fixed-design simulations retain actual times: 1,024 per design/scenario, seed 202611052. Errors are Gaussian iid variance 500 in working units; white SD 10 plus OU SD 20 with timescale 1 h, 12 h or 3 days; or white SD 10 plus independent period constants SD 30. Constants may be absorbed by nuisance columns and do not constitute a general drift test. Extended Data Tables 6 and 7 report ten-day whole-template and lunar performance. $\hat{A} \pm u_{\max}$ is a one-SE summary; normal and t_5 multiples define separate nominal 95% intervals. True-value wild acceptance is distinct from numerical inversion. The historical 124/200 injection coverage concerned one SE, not 95%, and cannot certify extreme-tail significance.

A Gaussian white-plus-OU diagnostic averages exponential covariance over actual member pairs, with

white-bin variance proportional to $600/n_i$. ML/REML profile a common scale over iid and 70 combinations of doubling timescales 10–5120 min and mixing fractions 0.1, 0.3, 0.5, 0.7, 0.9, 0.97 and 0.995. Whitened residuals show lag-one correlations −0.13 to −0.17, period RMS ratios 1.37–1.38 and kurtosis 4.82–5.12. None of 4,096 conditional simulations per selected model exceeds the observed maximum absolute lag-1–12 correlation, RMS ratio or kurtosis. They retain mean projection but do not reselect covariance parameters; these are not selection-adjusted probabilities. This noise family is inadequate for those diagnostics, and its likelihood significance is not used for detection.

Three further stress scenarios use variance-normalized Student-t7 or t5 innovations, negative MA(1) correlation −0.15, and fixed period scales from whole-template residuals. A third mixes t7 MA and member-averaged 12-h OU at variance fractions 0.9/0.1, adjusting MA correlation to approximately preserve lag one. MA memory restarts across interval changes or gaps ≥900 s. White-bin variance scales as $600/n_i$; period scales are $[18.518, 17.888, 19.313, 19.170, 24.699, 21.638] \times 10^{-18}$. Each has 4,096 simulations (seed 202611056), without re-estimating these empirical parameters or reselecting samples. Delete-day joint-phase regions cover 96.8–98.2%, versus 85.8–86.9% for delete-period ellipses and 96.8–96.9% wild acceptance. Joint wild power for unit response is 19.9–27.6%, versus 99.0–99.8% for the whole template without daily terms. These exploratory conditional tests neither identify instrument noise nor certify arbitrary systematics or model selection.

Additional short-period diagnostics

The whole-template point estimate is 0.8836; LS, delete-day and delete-period errors are 0.0882, 0.1002 and 0.0971. For the lunar scale 0.8909, they are 0.2344, 0.2824 and 0.1881. The ten-day block-drift change in the whole-template scale is −0.006 ± 0.030 under paired day deletions; five-day blocks with drift give 0.868 ± 0.112. The ten-day daily-term whole-template zero-response p-value is 0.03125, versus 0.0625 with five-day blocks. Fixed-phase lunar p-values are 0.0625 and 0.09375, respectively. All are conditional, two-sided period tests; only 64 sign patterns are available.

Across no local screening and the six-/eight-scale screens with five-/ten-day baselines, whole-template scales without daily terms range from 0.884 to 0.898 and lunar scales from 0.887 to 0.991. Removing local screening changes the ten-day daily-term whole-template p-value from 0.03125 to 0.0625. These are sensitivities on overlapping observations, not alternative independent detections.

The Figure 1a window is selected by retained source count, not residual agreement. This reproducible presentation rule was not prospectively registered before all exploratory analyses. Display averages contain 18 consecutive 600-s bins within uninterrupted supports; they do not bridge gaps. The six held-out comparisons share training observations.

Exposure and conditional sensitivity

Chronological prefixes first reach 2, 4, 8, 12 and 16 nominal sample days, plus full support, retaining complete bins (Extended Data Table 3). Thresholds preceded this calculation but followed full-data exploration. Nested prefixes change phase coverage, period composition and nuisance columns; they are not independent replications or sequential tests. LS, day and period errors are separate; fewer than three deletion groups are marked unavailable. Precision need not improve monotonically.

A conditional technical benchmark uses the measured summer 600-s clock Allan deviation, $\sigma_{600} = 2.3476\times10^{-17}$, as a white-frequency reference. Independent bin variances $v_i = \sigma_{600}^2(600/n_i)$ propagate through the unchanged OLS target operator O as $O\,\mathrm{diag}(v_i)O^T$. This does not reweight observations or certify independence and equal noise across periods. The full-support benchmark errors are 0.101 and 0.268 for whole-template and lunar scales. At four nominal days, the lunar benchmark error is 4.33 times that obtained by scaling the full-support error as $T^{-1/2}$, reflecting the different sampling and nuisance geometry. No Allan extrapolation to 20 days or universal detection time is inferred.

Spectral and held-out diagnostics

Harmonic scans generalize floating-mean least squares[23] using the nuisance projector $M_Z = I - ZZ^+$. At each frequency, the two-quadrature fit to $M_Z y$ gives residual-sum-of-squares reduction divided by sample size. Actual bin-mean times scan 0.25–4 cycles/day at $1/(5\Delta t)$ spacing, including specified constituents. Figure 1b subtracts $M_Z T$ with unit amplitude and fixed predicted phase. A separate archived-amplitude subtraction, $\hat{A}_T M_Z T$, leaves M2 diagnostics of 0.120×10^{-36} without daily terms

and 0.145×10^{-36} with them, corresponding to reductions of 99.1% and 93.8%. Both use identical observations and projection. Gaps remain intact; increases at other frequencies are retained in Extended Data Table 1. These scan ordinates mix sampled constituents and are not independent constituent-amplitude estimates. They are neither a physical PSD nor a calibrated frequency-search significance test; fixed-unit subtraction does not remove prior exploratory choices.

For leave-period-out diagnostics, five periods train the amplitude. The test response and template are projected against locally fitted nuisance columns before comparing residual sums of squares. The test amplitude is fixed; training observations overlap across folds.

Fibre monitor and stability

The 29 monitor files use the experimental B2 designation. Their nominal one-second BJT axes start at the floored first timestamp and advance by row index, matched to corresponding clock axes; repeated or nonfinite monitor seconds are excluded. Pairings at −5, 0 and +5 s test that convention without certifying synchronization or latency. All original members of a retained bin must match. Zero-shift unfiltered support has 915,039 samples, including 1,347 B2 deviations beyond 100 Hz. The post-inspection ±100-Hz excursion screen retains ±10/±1000-Hz alternatives and was not selected for tidal agreement.

We compare $y = A_T T + Z\eta + \epsilon$ with $y = A'_T T + Z\eta' + \lambda B2 + \epsilon'$. Paired day deletions estimate $A'_T - A_T$ uncertainty, retaining B2's own template projection. Frequency deviations use nominal optical carrier 193399200000000 Hz for a round-trip diagnostic; neither division by two nor 10-MHz RF normalization supplies a one-way optical error bound.

Overlapping Allan deviations[24] compare adjacent τ-means on contiguous common seconds, with no term across gaps and no run shorter than 4τ. Term counts are retained. Summer results are not extrapolated to other periods; timestamp checks do not certify hardware synchronization. No same-site null comparison is available.

For the full retained 2026 clock membership, nominal-second runs additionally split at file and effective-interval boundaries. The primary minimum length is 4τ; a separate 2τ rule is exploratory (Extended Data Table 4). At six hours, two runs from two periods give 4.35×10^{-18} before and 2.16×10^{-18} after subtraction of the fixed unit-amplitude prediction on identical support, without fitted amplitudes, phases, drifts or offsets. The latter is a model-conditional diagnostic, not an independent tidal test or instrument-noise bound. Eligible runs change with τ, so the curve is not solely an integration-gain law. The exploratory twelve-hour estimate has only seven overlapping differences. Overlap counts are not independent degrees of freedom, and no calibrated confidence interval is assigned. This full-record diagnostic does not replace the summer common-support benchmark used in the exposure analysis.

Long-period amplitudes and upper bounds

Long-period models jointly fit five quadrature pairs, short-period group scales and global daily terms on 2026 data with one intercept. Alternatives add one global drift, use component-sum corrections or omit May. No segment, period or month intercepts are used; all 2025 measurements are excluded.

The no-drift individual 95% amplitude bounds for Mf, Mm, Msf, Sa and Ssa are

$$(1.60, 4.32, 2.70, 20.98, 11.11) \times 10^{-17}$$

(Extended Data Fig. 1). A global drift expands Sa and Ssa bounds to approximately 6.0×10^{-15} and 1.1×10^{-15}. The no-drift Sa bound is about 75,000 times the current predicted amplitude. Neither the retained-level plotting coordinate nor these bounds represents the calibrated absolute-ratio coordinate of the earlier summer analysis.

For a pair $\hat{b}$ with covariance C and denominator degrees of freedom ν, a conservative radial bound from the nominal ellipse is

$$U = \|\hat{b}\| + \sqrt{2F_{2,\nu}(1-\alpha)\lambda_{\max}(C)}.$$

We use $\alpha = 0.05$ for an individual bound and 0.01 for the five-target Bonferroni family. With six period clusters, $\nu = 5$ is a working approximation. The no-drift family upper bounds for Mf, Mm, Msf, Sa

and Ssa are $(2.30, 6.18, 3.75, 30.38, 15.28) \times 10^{-17}$. Omitting May leaves five periods and gives individual Sa and Ssa bounds of approximately 1.37×10^{-15} and 9.36×10^{-16}. Bounds are conditional on the calibration coordinate; missing early Sr AC corrections and other unbounded changes prevent an unconditional gravitational limit.

The independent solar check averages the degree-two potential over hour angle. With P_2 the second Legendre polynomial, it evaluates $-(1 + k_2 - h_2)GM_\odot[r_W^2 P_2(\sin\varphi_W) - r_S^2 P_2(\sin\varphi_S)]P_2(\sin\delta)/(r_\odot^3 c^2)$ using DE430[20] and the archived theory-input coordinates (Wuhan: 30.529° N, 114.357° E, 50 m; Shanghai: 31.130° N, 121.570° E, 20 m), converted to geocentric radii and latitudes for this calculation. This checks the small Sa/Ssa scale without fitting observations. It does not provide a complete seasonal-loading model. The existing 18-constituent loading series contains none of Mf, Mm, Msf, Sa or Ssa.

Computational reproducibility

Independent acquisition is reported by the experimental team; new template, spectral and component analyses followed access to the European work and critique. Route, species and geography are not claimed as exclusive categories.

Versioned files retain source identities, exclusions, designs, covariances, seeds and figure inputs. An independent parser read all 1,737,073 retained 2026 records from 34 contributing original files, verifying hashes, row counts and identities. Direct 80-digit Decimal conversion and member averaging reproduce both ratio-coordinate routes within 8.2×10^{-28} fractional frequency. This verifies the frozen data path, not sample selection or experimental corrections. Independent QR/covariance checks agree. Principal calculations use Python 3.12.3, NumPy 2.5.4 and SciPy 1.18.1; astronomy uses NumPy 2.5.3 and pyTMD 3.0.9; figures use Matplotlib.

AI-assisted workflow

Analysis planning, implementation, numerical cross-checking and manuscript drafting used OpenAI Codex agents, including the recorded gpt-6.1-sol specialist model. Figures were rendered from numerical inputs using Python; simulations are identified separately from experimental observations. Specialist task instructions, code and execution records are retained. Separately implemented numerical checks within this workflow do not constitute independent human review. [The authors must verify this disclosure and the scientific content before submission.]

Data availability

A local referee bundle contains derived 600-s series, matched template columns and figure source data with file hashes. Source identities and exclusion records are retained locally. Authors must specify access and restrictions before submission; no public deposit is claimed.

Code availability

A portable entry point verifies the principal estimates and rebuilds the four main figures and Extended Data Fig. 1 from frozen derived inputs. Original-text readback still requires the source files. Versioned code and execution records are retained; authors must confirm access for editors, referees and readers. No public release is claimed.

Methods references

16. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). Source
17. LEGOS, NOVELTIS & CLS FES2022 (Finite Element Solution) Ocean Tide. *CNES data set* (2024). Source
18. Lyard, F. H., Allain, D. J., Cancet, M., Carrère, L. & Picot, N. FES2014 global ocean tide atlas: design and performance. *Ocean Science* **17**, 615–649 (2021). Source
19. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophysical Journal International* **120**, 287–311 (1995). Source
20. Folkner, W. M., Williams, J. G., Boggs, D. H., Park, R. S. & Kuchynka, P. The planetary and lunar ephemerides DE430 and DE431. *Interplanetary Network Progress Report* **42-196**, 1–81 (2014). Source

21. Efron, B. & Stein, C. The Jackknife Estimate of Variance. *The Annals of Statistics* **9**, 586–596 (1981). Source
22. Cameron, A. C., Gelbach, J. B. & Miller, D. L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Review of Economics and Statistics* **90**, 414–427 (2008). Source
23. Zechmeister, M. & Kürster, M. The generalised Lomb-Scargle periodogram: a new formalism for the floating-mean and Keplerian periodograms. *Astronomy & Astrophysics* **496**, 577–584 (2009). Source
24. Riley, W. J. Handbook of frequency stability analysis. *NIST Special Publication* **1065** (2008). Source

Acknowledgements

[To be completed by the authors.]

Funding

[Funding agencies and award identifiers require author confirmation.]

Author contributions

[To be completed for every author.]

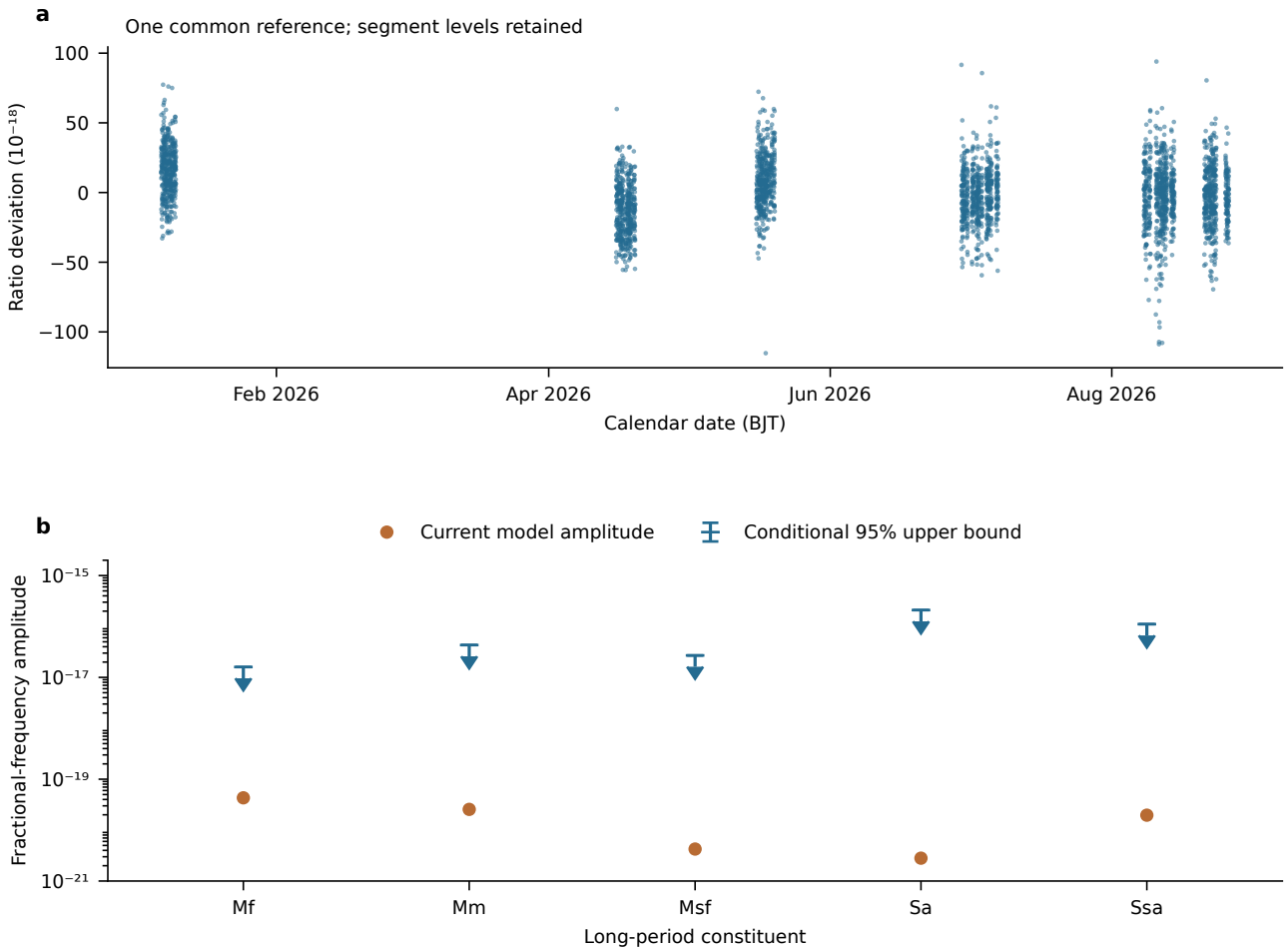
Competing interests

[Author declaration required; no absence of interests is presumed.]

Additional information

[Corresponding author and contact details require confirmation. All essential results, Methods and Extended Data are included in this document; no separate Supplementary Information is prepared.]

Extended Data



Extended Data Fig. 1 | Preserved frequency levels and long-period sensitivity. a, All 2,905 reconstructed 2026 ratio averages, with one common plotting reference subtracted. No fitted segment or period centres are removed. Points retain the actual gaps and period-to-period level differences. **b,** Nominal 95% upper bounds on free-phase amplitudes from the 2026-only, single-intercept model with joint short-period and global daily terms, without a global drift. Caps indicate upper bounds and arrows indicate the permitted lower-amplitude direction; arrow lengths are graphical. Bounds use the largest eigenvalue of each two-coefficient delete-period covariance and an approximate two-dimensional region with five denominator degrees of freedom. They are individual bounds, conditional on the stated calibration and noise assumptions. Orange points give amplitudes from the current theoretical decomposition, not measured signals. The five-constituent family bounds and drift sensitivity are specified in Methods. Unknown calibration terms and incomplete long-period loading prevent these bounds from being interpreted as unconditional gravitational limits.

Extended Data Table 1 | Fixed and estimated tide subtraction on identical support. Values are harmonic variance reduction per observation in 10^{-36}, evaluated at the six specified frequencies with the same nuisance projection before and after subtraction. Each of the four designs uses the same 2,905 observations. Unit subtraction fixes amplitude to one and phase to the prediction; fitted subtraction uses the archived whole-template amplitude for that design. These are correlated descriptive diagnostics, not independent tests or a physical PSD. A dash denotes projection degeneracy at exact S2 under daily controls. Rows with increased residual diagnostics are retained.

Baseline	Daily terms	Constituent	Before	Unit prediction	Fitted amplitude
5 days	No	O1	0.005279	0.920389	0.736122
5 days	No	K1	1.444382	0.359093	0.278646
5 days	No	N2	1.327991	0.146557	0.068003
5 days	No	M2	13.115935	0.070025	0.121481
5 days	No	S2	4.073222	0.078951	0.013937
5 days	No	K2	0.442990	0.390299	0.368992
5 days	Yes	O1	0.625955	0.211045	0.234686
5 days	Yes	K1	0.132705	0.059978	0.063349
5 days	Yes	N2	0.052519	0.203128	0.178780
5 days	Yes	M2	1.526034	0.000022	0.008892
5 days	Yes	S2	—	—	—
5 days	Yes	K2	0.091058	0.084483	0.063956
10 days	No	O1	0.005031	0.891094	0.706667
10 days	No	K1	1.408896	0.290857	0.207769
10 days	No	N2	1.300918	0.151914	0.071497
10 days	No	M2	13.020310	0.079641	0.120337
10 days	No	S2	4.034742	0.094409	0.018696
10 days	No	K2	0.433051	0.414359	0.389789
10 days	Yes	O1	0.296006	0.022191	0.044624
10 days	Yes	K1	0.356992	0.136847	0.143915
10 days	Yes	N2	0.027985	0.173398	0.086575
10 days	Yes	M2	2.333639	0.190457	0.145187
10 days	Yes	S2	—	—	—
10 days	Yes	K2	0.733570	0.227128	0.143159

Extended Data Table 2 | Joint lunar phase tests on the same observations. Six-period restricted wild tests compare the in-phase/quadrature pair with zero and with the unit, in-phase prediction. All 64 sign patterns are retained and each is studentized by its own delete-period covariance. These exploratory tests differ from fixed-phase scalar tests and the nominal delete-day ellipses in Fig. 2b. All designs use the same 2,905 observations; the primary ten-day baseline is unchanged.

Baseline	In phase	Quadrature	p, zero	p, unit in phase
5 days	0.892	-0.053	0.31250	0.90625
10 days	0.866	0.104	0.25000	0.71875
20 days	0.867	0.002	0.03125	0.62500
30 days	0.794	0.099	0.12500	0.25000
Observation period	0.754	0.067	0.03125	0.12500

Extended Data Table 3 | Accumulation on the actual 2026 record. Sample days sum retained members at a nominal one-second gate; span is elapsed BJT time. T and L denote the whole-template and daily-controlled lunar targets. **a,** Support and conditional independent-bin benchmark standard errors. **b,** Observed standard errors by method. "—" means fewer than three deletion groups; no missing error is assigned zero. These pointwise, nested comparisons do not have simultaneous confidence coverage or an optional-stopping guarantee.

a. Exposure, support and conditional benchmark

Sample days	Span (d)	Days / periods	T benchmark	L benchmark
2.003	2.2	3 / 1	0.362	2.985
4.007	99.9	6 / 2	0.241	2.599
8.001	131.7	12 / 3	0.143	0.911
12.005	179.6	20 / 4	0.123	0.424
16.004	217.8	28 / 5	0.107	0.314
20.105	231.3	35 / 6	0.101	0.268

b. Observed scale standard errors

Sample days	T LS	T day	T period	L LS	L day	L period
2.003	0.296	0.628	—	2.401	8.034	—
4.007	0.190	0.241	—	2.031	2.952	—
8.001	0.114	0.093	0.112	0.720	0.985	1.025
12.005	0.098	0.083	0.072	0.337	0.427	0.413
16.004	0.094	0.122	0.125	0.275	0.338	0.205
20.105	0.088	0.100	0.097	0.234	0.282	0.188

Extended Data Table 4 | Averaging-time support of the retained 2026 clock record. All 1,737,073 frozen members are retained before run-length requirements. Runs split at file and effective-interval boundaries or nonconsecutive nominal seconds. 4τ is the primary minimum run length; 2τ is an exploratory comparison, not a confidence level. Allan deviations are in 10^{-18} units; the last column subtracts the fixed unit prediction on identical support. Unsupported estimates are shown as —. Overlapping differences are not independent replicates. This support differs from the simultaneous B2 support in Fig. 4.

Minimum run	τ (s)	Runs / periods	Differences	Retained	Fixed tide subtracted
4τ	600	68 / 6	1,651,143	21.588	21.587
4τ	3,600	43 / 6	1,203,996	7.312	7.210
4τ	8,192	18 / 6	633,924	5.171	4.600
4τ	16,384	5 / 3	212,674	5.583	3.720
4τ	21,600	2 / 2	86,407	4.348	2.165
4τ	32,768	0 / 0	0	—	—
4τ	43,200	0 / 0	0	—	—
2τ	32,768	5 / 3	48,834	2.708	1.419
2τ	43,200	2 / 2	7	1.087	0.549

Extended Data Table 5 | Ten-day block support in 2026. Assignment uses the mean bin time; boundary-crossing source members remain intact. The start is inclusive and the end exclusive. All dates are BJT in 2026. Block duration describes the model, not continuous uptime.

Block	Calendar start–end	Bins	Source members	Observation period
2046	01-07–01-17	403	241,474	January
2055	04-07–04-17	181	108,008	April
2056	04-17–04-27	259	153,774	April
2058	05-07–05-17	121	72,085	May
2059	05-17–05-27	299	178,683	May
2063	06-26–07-06	527	315,603	June–July
2064	07-06–07-16	72	42,708	June–July
2067	08-05–08-15	592	355,036	August a
2068	08-15–08-25	341	203,702	August b
2069	08-25–09-04	110	66,000	August b

Extended Data Table 6 | Conditional simulation bias and standard errors. T: ten-day whole template without daily terms; L: M2+N2 with daily terms. Each Gaussian scenario uses 1,024 simulations; each Student/MA scenario uses 4,096 with fixed unequal period scales. Entries are bias and mean scale standard errors. These simulated responses do not certify arbitrary noise or selection effects. Corresponding interval coverage is in Extended Data Table 7.

Target / noise	Bias	LS SE	Day SE	Period SE
T / iid	-0.000	0.096	0.096	0.091
T / OU 1h	-0.007	0.094	0.264	0.255
T / OU 12h	-0.002	0.084	0.247	0.226
T / OU 3d	+0.003	0.064	0.146	0.134
T / period constants	+0.001	0.043	0.043	0.041
T / t7 MA	+0.002	0.088	0.074	0.071
T / t5 MA	+0.001	0.088	0.074	0.071
T / t7 MA + OU	+0.001	0.087	0.098	0.090
L / iid	-0.001	0.255	0.298	0.270
L / OU 1h	-0.002	0.238	0.799	0.714
L / OU 12h	+0.031	0.207	0.837	0.631
L / OU 3d	+0.004	0.164	0.594	0.359
L / period constants	+0.004	0.114	0.134	0.122
L / t7 MA	+0.004	0.236	0.244	0.213
L / t5 MA	+0.003	0.236	0.243	0.213
L / t7 MA + OU	+0.005	0.231	0.336	0.262

Extended Data Table 7 | Conditional interval coverage. Nominal 95% coverage (%) for the same targets and simulations as Extended Data Table 6. Columns report individual LS/day/period t intervals, the $1.96u_{\max}$ interval, and wild true-value acceptance. Gaussian and Student/MA scenarios have 1,024 and 4,096 simulations, respectively. True-value acceptance is distinct from numerically inverting a test, and none of these conditional scenarios provides a universal coverage guarantee.

Target / noise	LS	Day	Period	Max-SE	Wild
T / iid	95.0	94.8	94.2	96.7	96.5
T / OU 1h	49.9	94.1	93.7	94.5	96.5
T / OU 12h	49.7	95.6	94.1	96.1	97.4
T / OU 3d	60.9	96.4	94.0	95.9	96.9
T / period constants	94.7	93.9	93.8	96.0	96.0
T / t7 MA	97.9	94.6	94.6	98.3	96.5
T / t5 MA	98.4	95.2	94.7	98.9	96.7
T / t7 MA + OU	93.7	96.4	94.9	97.2	97.0
L / iid	95.2	96.8	94.4	98.3	96.9
L / OU 1h	50.2	96.3	93.1	96.9	96.7
L / OU 12h	49.9	99.0	92.9	98.8	96.5
L / OU 3d	67.0	99.8	93.5	99.8	95.8
L / period constants	96.2	97.5	94.1	98.3	97.5
L / t7 MA	96.9	96.7	92.7	98.2	96.6
L / t5 MA	97.1	97.0	92.4	98.3	96.7
L / t7 MA + OU	93.1	98.5	93.0	98.5	96.9